7.1

1) Write in terms of sin and cos. Simplify.

$\csc\theta \tan\theta$

$= \frac{1}{\sin\theta} \cdot \frac{\sin\theta}{\cos\theta}$

$= \frac{1}{\cos\theta}$

$= \boxed{\sec\theta}$

2) Write in terms of sin and cos. Simplify.

$8\sin^2\theta\,(1+\cot^2\theta)$

$= 8\sin^2\theta\,(\csc^2\theta)$

$= 8\sin^2\theta \cdot \frac{1}{\sin^2\theta}$

$= \boxed{8}$

3) Simplify

$\sin^3\left(\frac{\pi}{2}-x\right) + \sin^2 x \cos x$

$= \cos^3 x + \sin^2 x \cos x$

$= \cos x(\cos^2 x + \sin^2 x)$

$= \cos x \cdot 1$

$= \boxed{\cos x}$

4) Simplify

$\frac{6\csc^2 x - 6}{6\csc^2 x}$

$= \frac{\cancel{6}(\csc^2 x - 1)}{\cancel{6}(\csc^2 x)}$

$= \frac{\cot^2 x}{\csc^2 x}$

$= \boxed{\cos^2 x}$

5) verify the identity

$\frac{\cot x}{\csc x} = \cos x$

LHS: $\frac{\cot x}{\csc x}$

$= \frac{\frac{1}{\tan x}}{\frac{1}{\sin x}}$

$= \frac{\frac{\cos x}{\sin x}}{\frac{1}{\sin x}}$

$= \frac{\cos x}{1}$

$= \cos x$

$=$ RHS

6) Verify the identity.

$\sin(-x) - \cos(-x) = -\sin x - \cos x$

Sin is odd function

Cos is even function

LHS: $\sin(-x) - \cos(-x)$

$= -\sin x - \cos x$

$=$ RHS

7) Verify identity using fundamental trig identities

$(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$

Expan the square, then use a Pythagorean Identity

$(\sin x + \cos x)^2 = \sin^2 x + (2\sin x\cos x) + \cos^2 x$

$= 1 + (2\sin x \cos x)$

8) Verify identity using fundamental trig identities

$\frac{\sec x + \csc x}{\tan x + \cot x} = \sin x + \cos x$

$\frac{\sec x + \csc x}{\tan x + \cot x} = \frac{\frac{1}{\cos x} + \frac{1}{\sin x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$

$= \frac{\frac{1}{\cos x} + \frac{1}{\sin x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}} \cdot \frac{\sin x \cos x}{\sin x \cos x}$

$= \frac{\sin x \cos x}{\sin^2 x + \cos^2 x}$

$= \sin x + \cos x$

9) Make indicated trig substitution and simplify.

$\frac{x}{\sqrt{1-x^2}}, \quad x = \cos\theta$

$= \frac{\cos\theta}{\sqrt{1-\cos^2\theta}}$

$= \frac{\cos\theta}{\sqrt{\sin^2\theta}}$

$= \frac{\cos\theta}{\sin\theta}$

$= \boxed{\cot\theta}$

10) Simplify.

$\frac{1+\cos u}{\sin u} + \frac{\sin u}{1+\cos u}$

$= \frac{(1+\cos u)^2 + \sin^2 u}{(\sin u)(1+\cos u)}$

$= \frac{1 + 2\cos u + \overbrace{\cos^2 u + \sin^2 u}^{1}}{(\sin u)(1+\cos u)}$

$= \frac{2 + 2\cos u}{(\sin u)(1+\cos u)}$

$= \frac{2\cancel{(1+\cos u)}}{(\sin u)\cancel{(1+\cos u)}}$

$= \frac{2}{\sin u}$

$= \boxed{2\csc u}$

11) Simplify

$\frac{9\cot(A) - 9}{1+\tan(-A)}$

$= \frac{9(\cot A - 1)}{1-\tan A}$

$= \frac{9\left(\frac{1}{\tan A} - \frac{\tan A}{\tan A}\right)}{1-\tan A}$

$= \frac{\frac{9\cancel{(1-\tan A)}}{\tan A}}{\cancel{1-\tan A}}$

$= \frac{9}{\tan A}$

$= \boxed{9\cot A}$

12) Simplify.

$\frac{3}{1-\sin(\alpha)} + \frac{3}{1+\sin(\alpha)}$

$= \frac{3(1+\sin\alpha)}{(1-\sin\alpha)(1+\sin\alpha)} + \frac{3(1-\sin\alpha)}{(1+\sin\alpha)(1-\sin\alpha)}$

$= \frac{3(1+\sin\alpha) + 3(1-\sin\alpha)}{(1-\sin\alpha)(1+\sin\alpha)}$

$= \frac{3 \cancel{+3\sin\alpha} + 3 \cancel{-3\sin\alpha}}{1-\sin^2\alpha}$

$= \frac{6}{1-\sin^2\alpha}$

$= \frac{6}{\cos^2\alpha}$

$= \boxed{6\sec^2\alpha}$

13) Simplify

$\frac{2+\cot^2 x}{\csc^2 x} - 1$

$= \frac{2}{\csc^2 x} + \frac{\cot^2 x}{\csc^2 x} - 1$

$= 2\sin^2 x + \cos^2 x - 1$

$= \sin^2 x + \sin^2 x + \cos^2 x - 1$

$= \sin^2 x + 1 - 1$

$= \boxed{\sin^2 x}$

14) Simplify.

$\frac{5+6\cos y}{6+5\sec y}$

$$= \frac{5+6\cos y}{6+\frac{5}{\cos y}}$$

$$= \frac{5+6\cos y}{\frac{6\cos y+5}{\cos y}}$$

$$= \boxed{\cos y}$$

15) Simplify

$$\frac{9\csc x - 9\sin x}{9\cot x}$$

$$= \frac{\cancel{9}(\csc x - \sin x)}{\cancel{9}(\cot x)}$$

$$= \frac{\csc x - \sin x}{\cot x}$$

$$= \frac{\csc x}{\cot x} - \frac{\sin x}{\cot x}$$

$$= \sec x - \frac{\sin x}{\frac{\cos x}{\sin x}}$$

$$= \frac{1}{\cos x} - \frac{\sin^2 x}{\cos x}$$

$$= \frac{1-\sin^2 x}{\cos x}$$

$$= \frac{\cos^2 x}{\cos x}$$

$$= \boxed{\cos x}$$

16) Simplify

$$\frac{9\sin(t) + 6\tan(t)}{\tan t}$$

$$= \frac{9\sin t}{\tan t} + \frac{6\cancel{\tan t}}{\cancel{\tan t}}$$

$$= 9 \cdot \frac{\sin t}{\cos t} + 6$$

$$= \boxed{9\cos t + 6}$$

17) Write in terms of sin and cos. Simplify.

$$\frac{\tan\theta}{\sec\theta - \cos\theta}$$

$$= \frac{\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos\theta} - \cos\theta}$$

$$= \frac{\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos\theta} - \frac{\cos^2\theta}{\cos\theta}}$$

$$= \frac{\frac{\sin\theta}{\cancel{\cos\theta}}}{\frac{1-\cos^2\theta}{\cancel{\cos\theta}}}$$

$$= \frac{\sin\theta}{1-\cos^2\theta}$$

$$= \frac{\sin\theta}{\sin^2\theta}$$

$$= \frac{1}{\sin\theta}$$

$$= \boxed{\csc\theta}$$

18) Write in terms of sin and cos. Simplify.

$$\sin u \tan u + \cos u$$

$$= \sin u \frac{\sin u}{\cos u} + \cos u$$

$$= \frac{\sin^2 u}{\cos u} + \frac{\cos^2 u}{\cos u}$$

$$= \frac{\sin^2 u + \cos^2 u}{\cos u}$$

$$= \frac{1}{\cos u}$$

$= \boxed{\sec u}$

19) Write in terms of sin and cos. Simplify.

$\tan\left(\frac{\pi}{2}-u\right)\sin u$

$= \cot u \sin u$

$= \frac{\cos u}{\sin u}\sin u$

$\boxed{= \cos u}$

20) Write in terms of sin and cos. Simplify.

$\sin^2\left(\frac{\pi}{2}-y\right)\sec(y)$

$= \cos^2 y \sec y$

$= \cos^2 y \cdot \frac{1}{\cos y}$

$= \cos y$

21) Write in terms of sin and cos. Simplify.

$\cot^2 x - \csc^2 x$

$= \frac{\cos^2 x}{\sin^2 x} - \frac{1}{\sin^2 x}$

$= \frac{\cos^2 x - 1}{\sin^2 x}$

$= \frac{-\sin^2 x}{\sin^2 x}$

$= \boxed{-1}$

22) Simplify.

$\frac{1+\cos\left(y-\frac{\pi}{2}\right)}{1+\csc y}$

$= \frac{1+\sin y}{1+\csc y}$

$= \frac{1+\sin y}{1+\frac{1}{\sin y}}$

$= \frac{1+\sin y}{\frac{1+\sin y}{\sin y}}$

$= \frac{\sin y + \sin^2 y}{1+\sin y}$

$= \frac{\sin y\cancel{(1+\sin y)}}{\cancel{1+\sin y}}$

$= \boxed{\sin y}$

23) Verify the identity using trig identities

$\frac{\cos(\alpha)}{\sec(\alpha)} = \cos^2\alpha$

Use Reciprocal Identity to rewrite and then simplify

$\frac{\cos(\alpha)}{\sec(\alpha)} = \frac{\cos(\alpha)}{\left(\frac{1}{\cos(\alpha)}\right)}$

$= \cos(\alpha)\cdot(\cos(\alpha))$

$= \cos^2\alpha$

24) Verify the identity using trig identities.

$\cos^2\left(\frac{\pi}{2}-y\right)\csc y = \sin y$

Select the correct identities and algebraic manipulations that justify each step in the proof below

$\cos^2\left(\frac{\pi}{2}-y\right) = \sin^2 y \csc y$ Confunction Identity

$= \sin^2 y \cdot \frac{1}{\sin y}$ Reciprocal Identity

$= \sin y$ Cancel Common factor

25) Verify the identity using trig identities.

$\frac{1}{1-\sin^2 y} = 1+\tan^2 y$

Use a Pythagorean Identity in the denominator, and then use a Reciprocal Identity

$\frac{1}{1-\sin^2 y} = \frac{1}{\cos^2 y}$

$= \sec^2 y$

Use Pythagorean Idntity

$= 1 + \tan^2 y$

26) Verify the identity using trig identities.

$(\tan x + \cot x)^2 = \sec^2 x + \csc x$

Expand the product and use the Reciprocal and Pythagorean Identities to simplify

$$(\tan x + \cot x)^2 = \tan^2 x + 2\tan x \cdot \cot x + \cot^2 x$$

$$= \tan^2 x + 2 + \cot^2 x$$

$$= (\tan^2 x + 1) + (\cot^2 x + 1)$$

$$= \sec^2 x + \csc^2 x$$